

Alejandro Maldonado Medina

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Professional summary

Bachelor of Science in Geosciences from UNAM with a specialization in Applied Geophysics. My undergraduate thesis focused on a 3D electrical resistivity tomography study of the central courtyard of the Palacio de Minería in Mexico City, providing valuable insights into the building's structural stability and its differential subsidence. I have a strong interest in geophysical exploration for urban and archaeological settings, as well as applications in the energy sector, particularly in the development and sustainable use of geothermal energy. I complement my academic background with studies in data science, focused on data analysis, visualization, and database management using Python and SQL. I apply these skills in interdisciplinary projects that integrate diverse approaches to solve complex problems through data-driven solutions. In addition, I develop and maintain a personal blog where I share my academic and personal projects related to geosciences and data science: <https://alexmm28.github.io/>

Education

Escuela Nacional de Estudios Superiores Unidad Morelia. UNAM.

Bachelor's degree in Geosciences with the área of Applied Geophysics August 2019 – June 2023

Bachelor's thesis: Tomografía de resistividad eléctrica 3D en el palacio de Minería, Ciudad de México

Bachillerato Gestalt

Chemical-biological baccalaurate

August 2016 – June 2019

Professional experience

Institute of Geophysics, Michoacan Unit

Intern/Social service

March 2022 – September 2022

Creation of an interactive map as part of the P02 project, which consisted of a map of geothermal regions in Mexico based on fluid geochemistry and aquifer distribution through the digitalization of databases to a web page.

Courses, certifications and volunteering

- **Fiesta de la Ciencias y las humanidades** October 2019
UNAM Campus Morelia. Expositor del "Calendario Cósmico".
- **Basic level seismic data analysis for Institute of Geophysics, UNAM** January 2023
To verify this certification, please consult: <https://sigeco.cuaieed.unam.mx/valida.php> entity: IGEF and the folio number : 1429
- **Advanced level seismic data analysis for Institute of Geophysics, UNAM** January 2023
To verify this certification, please consult: <https://sigeco.cuaieed.unam.mx/valida.php> entity: IGEF and the folio number : 1370
- **Segundo Encuentro de Estudiantes de Geociencias** May, 2023
UNAM Campus Morelia. Organizing Committee. Scientific Poster Presenter
- **Analyze Data with Python Skill Path** April 2025
Data analysis course with statistics using Python and libraries such as Matplotlib, NumPy, and SciPy.
To verify this certification, visit:
<https://www.codecademy.com/profiles/alejamm28/certificates/a91cf883da2f48e2b977577646801c81>

- **How to Clean Data with Python** May 2025
Course to learn the basics of regular expressions and how to extract and clean data from the web with Python.
To verify this certification, visit:
<https://www.codecademy.com/profiles/alejamm28/certificates/e773a003314c1be60da8388a90a77e78>
- **Data Science Foundations** May 2025
Course to clean, analyze, and visualize data with Python and SQL
To verify this certification, visit:
<https://www.codecademy.com/profiles/alejamm28/certificates/738a7262ff2b4a49b062c25482dc2549>
- **6^a Virtual Conference of Latin American students of geophysics – SEG (Society of Exploration Geophysicists).** October 2025
- Presentation in English of “3D Electrical Resistivity Tomography at the Palace of Mining, Mexico City” at a specialized academic forum, highlighting geophysical applications and results. Winner of 2nd place.
- **Data Scientist: Machine Learning** In progress
Data Science programming course in Machine Learning to build models and generate meaning from data on the Codecademy platform.

Technical skills

Competencies: Data analysis, reading and interpretation of scientific articles in English, fieldwork, handling and use of geophysical equipment and techniques, interpretation of geophysical data, creation of 2D and 3D models, management of GIS, custom graphics with matplotlib and seaborn, linear regression, logistic regression, correlation analysis, handling of categorical variables (one-hot encoding), model evaluation (accuracy, precision, recall, F1-score), cleaning, transformation and scaling of variables (StandardScaler, MinMaxScaler).

Soft skills: Interdisciplinary teamwork, assertive communication, problem-solving, empathy, organization, leadership, adaptability, flexibility, initiative, critical and analytical thinking.

Idioms: Spanish (Native), English (advanced C1), Italian (basic A1).

Software: SEISAN earthquake analysis software, EarthImager 2D, EarthImager3D, OasisMontaj, Python, NumPy, Pandas, matplotlib, seaborn, scikit-learn, SQL, Jupyter Notebooks, Google Colab ArcGIS, QGIS, Microsoft Office, Google Drive, Google Earth Pro.